

# Isaac Mubiru

isaacmubiru99@gmail.com | [linkedin.com/in/isaac-mubiru-3bb728174](https://www.linkedin.com/in/isaac-mubiru-3bb728174) | [github.com/zac-09](https://github.com/zac-09) | Kampala, Uganda (UTC+3 — full EMEA overlap)

## PROFESSIONAL SUMMARY

Senior engineer who owns the platform layer: Kubernetes, AWS and GCP, CI/CD, and cloud economics. Led the zero-downtime migration of 20+ production apps at MTailor — architecture, rollback paths, and the cutover playbook end to end — and cut cloud spend by \$5,000/month by consolidating the full AWS surface area. Ran production Kubernetes and Docker microservices, built developer tooling and documentation that let teams self-serve, and use modern AI tools daily to debug, document, and reduce operational toil (see [github.com/zac-09](https://github.com/zac-09)). Fully remote since 2020 across US, Singapore, and Uganda time zones — async-first, clear in writing, generous with context.

## CORE COMPETENCIES

Kubernetes & Docker

AWS & GCP Infrastructure

CI/CD & Rollback Design

Cloud Cost Optimization

Node.js & TypeScript

Python

Developer Tooling & Docs

AI-Assisted Operations

## WORK EXPERIENCE

### MTailor

Sep 2022 — Present

Software Engineer — US Remote · Reporting directly to the CTO

- **Designed the zero-downtime cutover system for 20+ production apps:** real-time two-way MongoDB↔Firestore sync (Node.js + Google Pub/Sub) with per-service instant rollback — progressive-delivery discipline applied to a live platform migration
- **Cut cloud spend \$5,000/month** by auditing and consolidating the full AWS surface area (S3, EC2, EBS) onto GCP — rightsizing, deleting, and migrating every load-bearing service
- Migrated file storage from Amazon S3 to Google Cloud Storage with automated Python tooling
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping, generating an additional \$40 revenue per order

### MTailor

Mar 2022 — Jul 2022

Contractor (Backend Migration) — US Remote

- Built the migration pipeline reading from MongoDB and writing to Firestore with complex transform logic
- **Wrote developer documentation** for the new Firebase SDKs (Node.js and Python) that let the engineering team self-serve post-migration
- **Trained the Ops team** on the new Firebase dashboard and backup procedures — documentation and runbooks that held up without me in the room

### Dr Wealth

Aug 2021 — Jan 2022

Software Engineer — Singapore Remote

- Built batch jobs querying 2M+ Firestore records with cursor pagination and selective updates — kept stock prices current from Morningstar APIs within a fixed read budget
- Extended backend APIs with Firebase Cloud Functions and Express.js on Heroku

### CodeBits

Jan 2020 — Jul 2021

Software Engineer (Team Lead) — Uganda

- **Ran production Kubernetes** for an NGO case-management platform: containerized microservices (Docker), Apache Kafka event backbone, deployment and scaling ownership
- Led a team of 4 developers building legal-tech systems for FIDA Uganda and LASPNET
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC
- Shipped a cross-platform mobile app (React Native, Expo) with Firebase push notifications

## PROJECTS

---

### Zero-Downtime Platform Migration (MTailor)

Migrated 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime over a multi-year parallel run — service-by-service cutover, each with an instant rollback path.

Node.js · GCP Pub/Sub · MongoDB · Firestore · Python

### Real-Time Two-Way Sync Pipeline

Live MongoDB↔Firestore synchronization supporting parallel production traffic during migration — eventual consistency, idempotent writes, and observability around every message.

Node.js · Google Pub/Sub · Firestore · MongoDB

### Career-Ops — AI-Assisted Automation Pipeline (open source)

Agentic pipeline built on Claude Code that automates repetitive evaluation workflows end-to-end: browser-verified data extraction (Playwright), batch workers, and PDF generation — AI used to reduce toil, with a ground-truth verification layer.

Node.js · Playwright · Claude Code · YAML

## EDUCATION

---

BSc Computer Engineering — Makerere University

2018 – 2022

## SKILLS

---

**Proficient:** Node.js, TypeScript/JavaScript, Kubernetes, Docker, GCP, Firebase, Python, MongoDB, SQL, Git, Linux, React

**Intermediate:** AWS, Go, Rust, Apache Kafka, gRPC, React Native    **Languages:** English, Kiswahili